

LupiScan

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Abstract

This project presents a low-cost, AI-powered system for early detection of lupus (Systemic Lupus Erythematosus) through facial analysis. Leveraging computer vision and deep learning, the system utilizes a convolutional neural network (CNN) model trained to identify visual indicators of lupus from facial images. The model was developed using the YOLOv8 object detection architecture and trained via the Roboflow platform with an augmented dataset for enhanced accuracy. This solution aims to assist in the early screening of lupus, particularly in remote or resource-limited environments where access to healthcare is restricted.

Introduction

Systemic Lupus Erythematosus (SLE), commonly known as lupus, is a chronic autoimmune disease that can affect various parts of the body, including the skin, joints, kidneys, and other organs. Early diagnosis is crucial for managing symptoms and preventing severe complications, but it remains a challenge, especially in remote or underserved areas where access to healthcare specialists is limited. Traditional diagnostic procedures often involve multiple clinical tests and specialist evaluations, which may not be readily available to all patients.

Methodology

This project involves several integrated stages combining data preparation, model training, and embedded deployment. Initially, a dataset of facial images showing symptoms of lupus, particularly the butterfly rash, was collected and augmented using Roboflow, increasing the dataset from 106 to 610 images to improve model robustness. The YOLOv8 object detection algorithm was then employed to train a model to identify lupus-related features in facial images. Annotations were manually added to define areas of interest. The system was evaluated based on detection accuracy, speed, and usability, demonstrating its potential as a practical tool for early lupus screening in low-resource environments.

Accurately Detect the Butterfly-Shaped Rash on The Face.

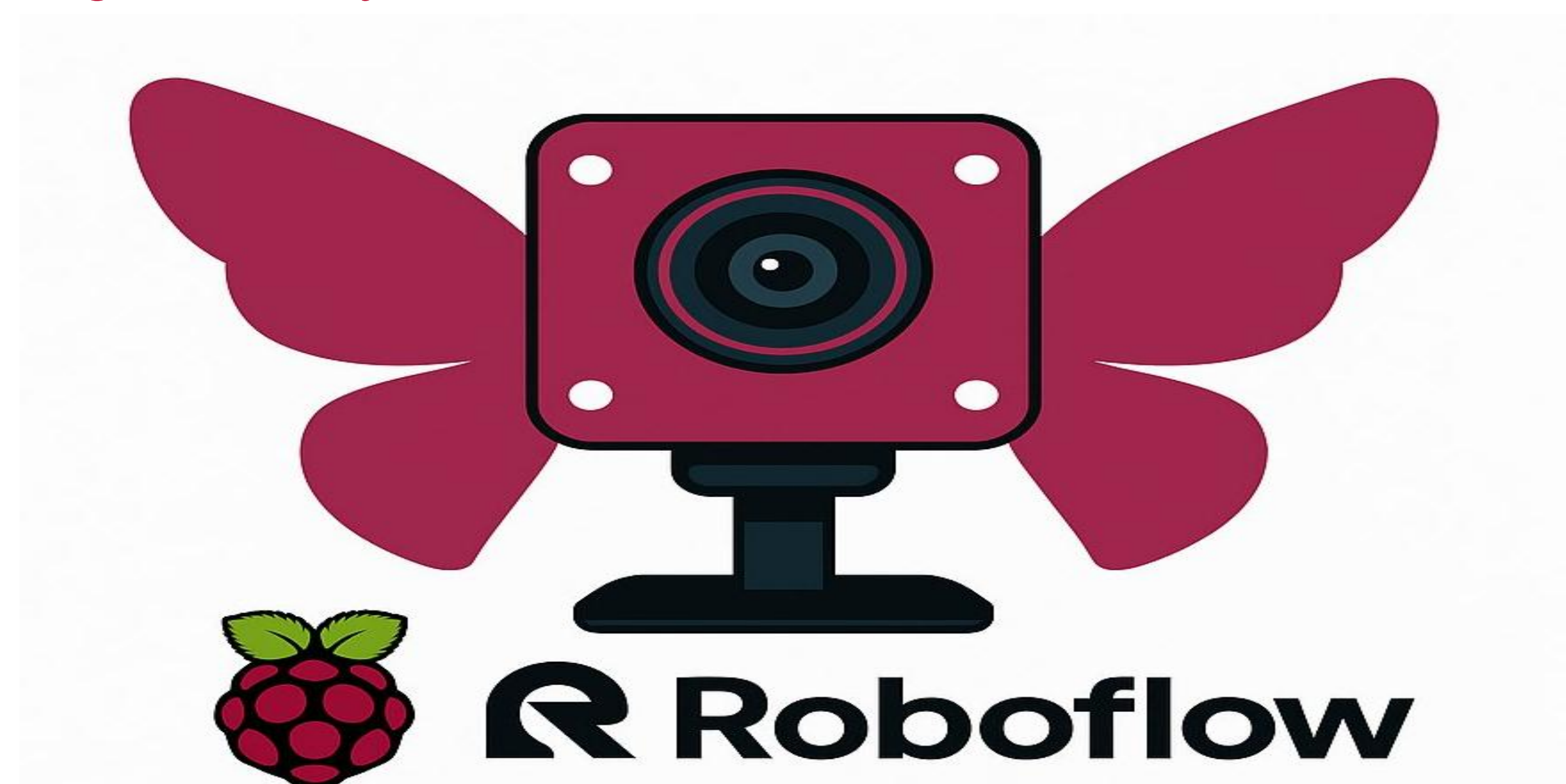


Results

The developed lupus detection system yielded encouraging results in both performance and usability. The YOLOv8 model, trained on an augmented dataset of 610 facial images, demonstrated strong object detection capabilities with an estimated mean Average Precision (mAP) of [84.2%] and high precision and recall values, indicating reliable identification of lupus-related facial features. The system was tested under varying conditions, including different lighting environments and facial angles, and continued to deliver consistent and accurate results.

Additionally, the model proved to be lightweight enough to run efficiently on the limited hardware resources of the Raspberry Pi, without the need for cloud-based processing. These results support the viability of using edge AI for early screening of lupus, particularly in remote or under-resourced medical settings where quick and cost-effective diagnostics are critical.

Logo of The Project



Discussions

The results of this project highlight the potential of integrating deep learning with embedded systems to support early disease detection, specifically for lupus. By utilizing YOLOv8 for object detection and deploying the model on a Raspberry Pi 4, we demonstrated that accurate, real-time inference can be achieved even on low-power devices. The ability to detect facial symptoms without relying on cloud infrastructure makes the system highly suitable for use in rural or resource-limited areas. However, several challenges were encountered, including the limited size and variability of the dataset, which may impact the generalizability of the model across different skin tones, facial structures, and lighting conditions. Furthermore, the detection relies solely on visual cues, which may not be sufficient for a definitive diagnosis, as lupus is a complex disease that often requires multiple tests and clinical evaluations. Despite these limitations, the system offers a promising supplementary tool for preliminary screening and awareness. Future improvements may include expanding the dataset with more diverse images, incorporating multi-modal data (e.g., patient history or symptom questionnaires), and enhancing the mobile application interface for broader usability.



Tafila Technical University

Accurately Detect the Butterfly-Shaped Rash for The Face.



Future Work

To enhance the effectiveness and scalability of the proposed lupus detection system, several areas of improvement are planned for future work. First, expanding the dataset to include a larger and more diverse set of facial images across different age groups, skin tones, and stages of the disease will improve the model's accuracy and generalizability. Second, integrating additional diagnostic indicators—such as patient-reported symptoms, medical history, or other visible skin conditions—could improve the system's reliability beyond visual detection alone. Third, optimizing the mobile application to include features like user authentication, result history tracking, and integration with electronic health records would make the system more practical for clinical use. Moreover, implementing real-time data synchronization and remote physician review features would support telemedicine applications. Finally, exploring the use of other lightweight deep learning models and hardware acceleration (e.g., Coral USB Accelerator or NVIDIA Jetson Nano) could significantly improve inference speed and efficiency for more complex deployments.

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